Filippo Lollato

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Education

M.Sc. in Artificial Intelligence Systems *2024/25 - Present*
University of Trento *Trento, Italy*

- **Focus Areas**: AI Systems Integration, Model Optimization & Fine-Tuning, Multi-Agent Systems, Deep Learning, Scalable Software Architecture.
- **Objective**: Actively seeking an industry Master's Thesis / Research Internship at an AI company/lab to optimize and integrate intelligent models into real-world services and production workflows.

B.Sc. in Computer Science *2021/22 - 2023/24*
University of Trento *Trento, Italy*

- **Bachelor's Thesis**: *Kindle: A Machine Learning Framework for Compile-Time Invariant and Dimensionality Safety* (Supervisor: Prof. Marco Patrignani).
- **Key Coursework**: Advanced Algorithms & Data Structures, Systems Programming, Computer Architecture, Operating Systems, Linear Algebra, Probability & Statistics.

Experience

M.Sc. Student in Artificial Intelligence Systems *2024/25 - Present*
University of Trento *Trento, Italy*

- MSc student focusing on model optimization, multi-agent systems, and integrating AI models into reliable software systems.
- Lead systems architect on the ASA Autobots multi-agent capstone project.

Teaching Assistant / Academic Tutor *2023/24 - Present*
University of Trento *Trento, Italy*

- Tutoring undergraduate students in Rust, Advanced Programming, and Compilers since 2023, beginning in the third year of my BSc in Computer Science.

Featured Projects

incin: Type-Safe Deep Learning Engine *Rust, CUDA, Const Generics*
High-assurance machine learning framework enforcing tensor invariants at compile time.

- Architected a high-performance deep learning engine in Rust utilizing affine type-level invariants and const generics to guarantee tensor shape and rank compatibility at compile time with zero runtime overhead.
- Engineered automatic differentiation graph execution, Conv2d operators with CUDA gradient parity verification, optimizer checkpointing, and modular data transformations.

ASA Autobots: Hybrid LLM & Symbolic Multi-Agent System *Node.js, PDDL, BDI, Local LLM Integration*
Real-time cooperative multi-agent logistics platform integrating LLM reasoning with symbolic planning.

- Engineered a distributed multi-agent system integrating BDI autonomous agents, local LLM reasoning (Llama 3.3 70B), and PDDL symbolic planners into real-time delivery logistics.
- Implemented structured JSON-RPC / tool-calling protocols, resilient error handling, and robust state synchronization with ~91.5% integration test coverage across a 3-layer architecture.

signal_image_video: 3D Multi-Object Tracking via Sensor Fusion *Python, PyTorch, LiDAR, EKF, KITTI*
Multi-modal camera-LiDAR fusion tracking system evaluated on the KITTI benchmark.

- Designed a 3D LiDAR-camera multi-modal sensor fusion pipeline combining Constant Turn Rate & Velocity Extended Kalman Filtering (CTRV-EKF) with Hungarian bipartite assignment.
- Evaluated on the autonomous driving KITTI benchmark, achieving superior trajectory tracking stability and measurable RMSE reduction over linear baselines.

NLU: Joint Intent Classification & Slot Filling *Python, PyTorch, Transformers, BERT*
End-to-end natural language understanding pipeline evaluated on the ATIS benchmark.

- Built an end-to-end multi-task NLU pipeline fine-tuning transformer architectures (BERT) with joint intent classification and sequence labeling heads from scratch.
- Attained 0.9765 intent classification accuracy and state-of-the-art slot-filling F1 score on the ATIS benchmark.

evolcpp: Systems Neuroevolution Engine *C++20, Template Metaprogramming, CMake*
Modern C++ implementation of compile-time verified neural abstractions.

- Implemented core tensor abstractions using advanced C++20 template metaprogramming to replicate compile-time shape verification and tensor safety without dynamic memory overhead.

Technical Skills & Certifications

Model Optimization & Tuning: Fine-Tuning (LoRA, QLoRA), Model Distillation, Quantization (NVFP4, GGUF, AWQ, FP8), Alignment (DPO, RLHF), Multi-Task Learning, Model Pruning, HuggingFace

AI Systems & Service Integration: Multi-Agent Systems (BDI, PDDL), Tool Use & Function Calling, LLM Orchestration, Model Serving (vLLM, Ollama, ONNX Runtime), Streaming APIs (SSE, WebSockets), RAG

Deep Learning & Perception: PyTorch, Transformer Architectures (BERT), Diffusion Pipelines, 3D LiDAR-Camera Sensor Fusion, Kalman Filtering (CTRV-EKF), CUDA C/C++

Languages & Backend Systems: Rust, C++ (C++20), Python, TypeScript, Linux (Debian/Ubuntu), Docker, CMake, Cargo, REST APIs, Axum, Node.js, Git, CI/CD

Certifications & Languages: Cambridge English: Level C1 (Pass at Grade A, Score 182) • Italian (Native)